

Update

- We have considered
  - Analog modulation techniques
  - Amplitude modulation
    - DSB-LC, DSB-SC, QAM, SSB, VSB
  - Frequency division multiplexing (FDM)
- We now consider
  - **Angle modulation**
    - FM, PM

Angle Modulation

A continuous-wave (CW) sinusoidal signal can be varied by changing its amplitude and its phase angle.

$$s(t) = A(t)\cos[\omega_c t + \phi(t)]$$

To carry a message signal $x(t)$:

Amplitude modulation:

Keep $\theta(t)$ constant and varies $A(t)$ proportionally to $x(t)$.

$$s(t) = A(t)\cos[\omega_c t + \phi_o] \quad where \;\; A(t) \propto x(t)$$

Angle modulation:

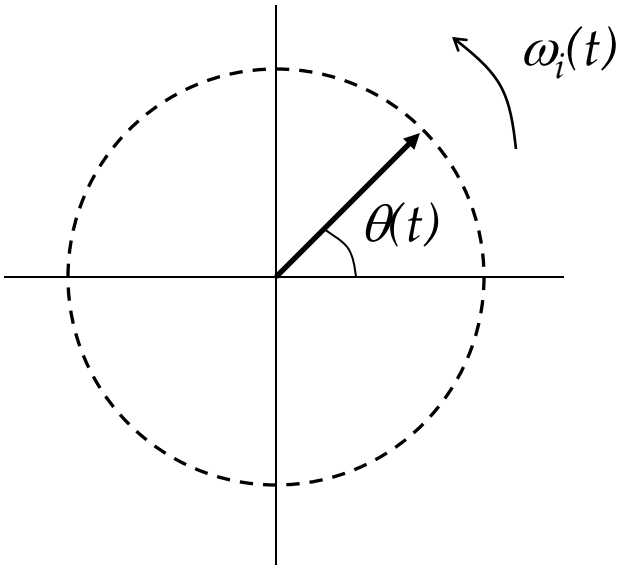
Keep $A(t)$ **constant** and varies $[\omega_c t + \phi(t)]$ **proportionally** to $x(t)$.

$$s(t) = A\cos[\omega_c t + \phi(t)] \quad where \;\; [\omega_c t + \phi(t)] \propto x(t)$$

Angle Modulation

Phasor Representation

The **phasor representation** of a constant-amplitude sinusoid is shown as follows



A: **magnitude of the phasor**

$\theta(t)$: **phase angle**

$$\theta(t) = \omega_c t + \phi(t) + \theta_o$$

$\omega_i(t)$: **instantaneous angular rate**

$$\omega_i(t) = \frac{d\theta(t)}{dt} = \omega_c + \frac{d\phi(t)}{dt}$$

$$\theta(t) = \int_0^t \omega_i(\tau)d\tau + \theta_0$$

Angle Modulation

If the phase $\phi(t)$ is varied linearly with the input signal $x(t)$, we have

$$\theta(t) = \omega_c t + k_p x(t) + \theta_0 \qquad i.e.\ \phi(t) \propto x(t)$$

where ω_c , k_p , θ_0 are constants.

As the **phase is linearly related to** $x(t)$, this type of angle modulation is called ***phase modulation*** **(PM)** with

$$s_{PM}(t) = A_c \cos[\omega_c t + k_p x(t) + \theta_o]$$

and , $\phi(t)$ **is called instantaneous phase deviation.**

$$\omega_i(t) = \frac{d\theta(t)}{dt} = \omega_c + \frac{d\phi(t)}{dt} \Rightarrow \omega_{i(PM)} = \omega_c + k_p \frac{dx(t)}{dt}$$

Angle Modulation

If the instantaneous frequency ω_i proportional to the input signal, we have

$$\omega_{i(FM)}(t) = \omega_c + k_f x(t) \Rightarrow \frac{d\phi(t)}{dt} \propto x(t) \qquad \because \omega_i(t) = \frac{d\theta(t)}{dt} = \omega_c + \frac{d\phi(t)}{dt}$$

$$\theta(t) = \omega_c t + k_f \int_0^t x(\tau) d\tau + \theta_o$$

where ω_c, k_f are constants.

As the frequency is linearly related to $x(t)$, this type of angular modulation is called *frequency modulation* (FM) , with

$$s_{FM}(t) = A_c \cos[\omega_c t + k_f \int_0^t x(\tau) d\tau + \theta_o]$$

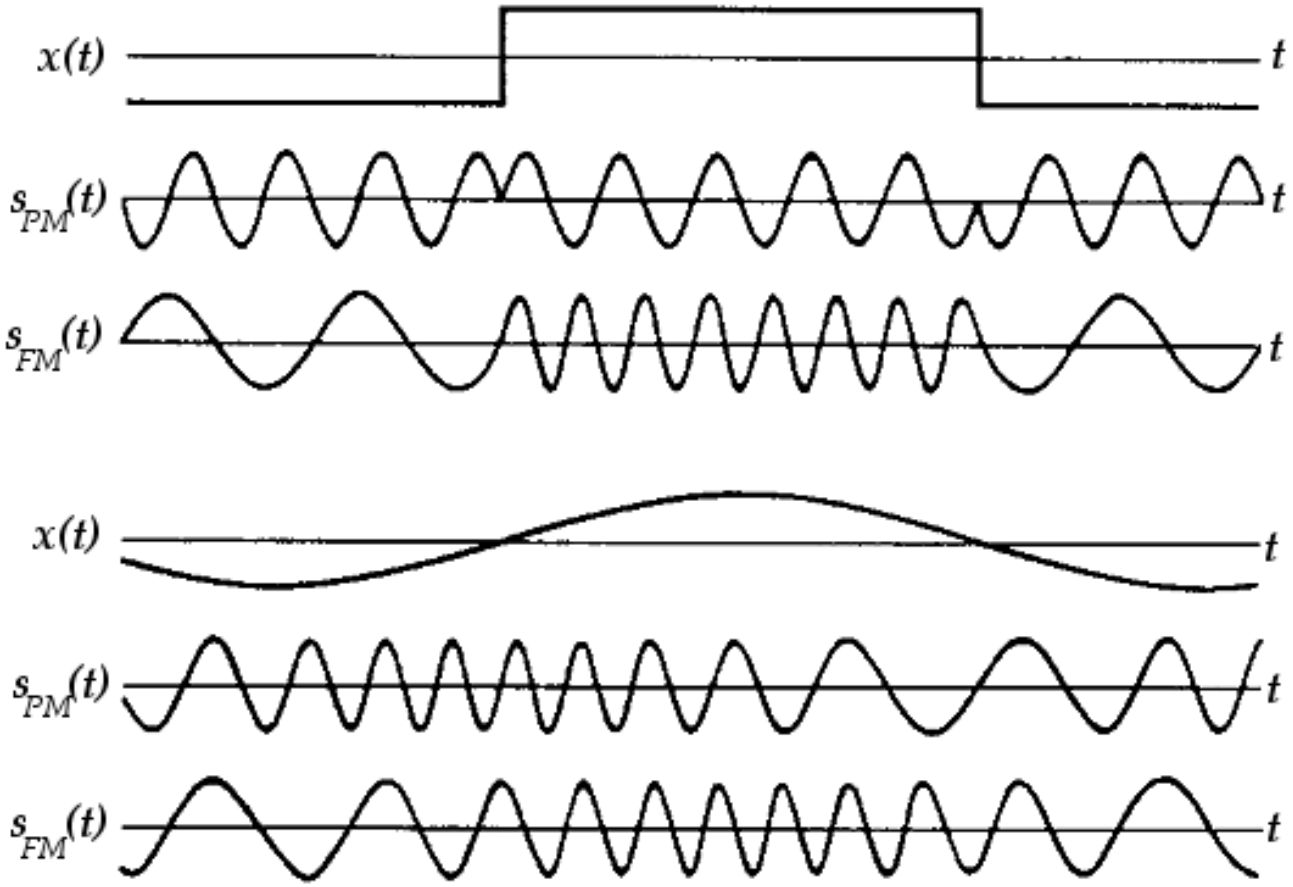
and; $\frac{d\phi(t)}{dt}$ is called instantaneous frequency deviation.

Angle Modulation

	Phase Modulation (PM)	Frequency Modulation (FM)
Instantaneous angular rate	$\omega_{i(PM)} = \omega_c + k_p \frac{dx(t)}{dt}$	$\omega_{i(FM)}(t) = \omega_c + k_f x(t)$
Modulated signal	$s_{PM}(t) = A_c \cos[\omega_c t + k_p x(t) + \theta_o]$	$s_{FM}(t) = A_c \cos[\omega_c t + k_f \int_0^t x(\tau) d\tau + \theta_o]$
Proportionality	$\phi(t) \propto x(t)$	$\frac{d\phi(t)}{dt} \propto x(t)$

Angle Modulation

FM and PM Waveforms



Angle Modulation: Fourier spectra

Consider an **angle-modulated signal**

$$\begin{aligned} s(t) &= A_c \cos[\omega_c t + \phi(t)] \\ &= Re\left\{A_c e^{j[\omega_c t + \phi(t)]}\right\} \\ &= Re\left\{A_c e^{j\omega_c t} e^{j\phi(t)}\right\} \end{aligned}$$

Expand $e^{j\phi(t)}$ in a power series, gives

$$s(t) = \mathrm{Re}\left\{A_c e^{j\omega_c t}\left[1 + j\phi(t) - \frac{\phi^2(t)}{2!} - \cdots + j^n \frac{\phi^n(t)}{n!} + \cdots\right]\right\}$$

$$= A_c\left[\cos\omega_c t - \phi(t)\sin\omega_c t - \frac{\phi^2(t)}{2!}\cos\omega_c t + \frac{\phi^3(t)}{3!}\sin\omega_c t + \cdots\right]$$

The signal consists of an unmodulated carrier plus **various amplitude-modulated terms**, such as $\phi(t)\sin\omega_c t$, $\phi^2(t)\cos\omega_c t$, $\phi^3(t)\sin\omega_c t$, …, etc.

Hence the Fourier spectrum consists of an unmodulated carrier plus spectra (sidebands) of $\phi(t)$, $\phi^2(t)$, $\phi^3(t)$, …, etc., centered at ω_c.

Angle Modulation: Narrowband

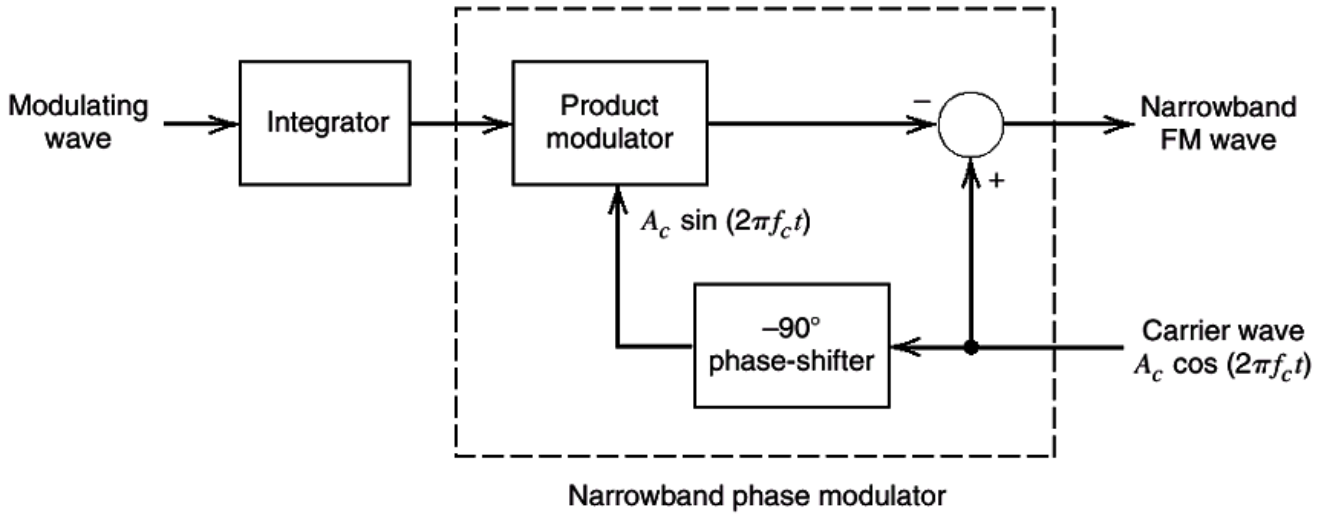
If $|\phi(t)|_{max} << 1$, then, by neglecting the higher-power terms of $\phi(t)$ in the $s(t)$, gives

$$s(t) \approx A_c \cos\omega_c t - A_c \phi(t) \sin \omega_c t$$

which is called the ***narrowband (NB) angle-modulated signal***.

$$s_{NBPM}(t) \approx A_c \cos\omega_c t - A_c k_p x(t) \sin \omega_c t$$

$$s_{NBFM}(t) \approx A_c \cos\omega_c t - A_c \left[k_f \int_0^t x(\tau) d\tau \right] \sin \omega_c t$$



Angle Modulation: NBFM (Sinusoid)

Narrowband FM (NBFM)

Consider $x(t) = a_m \cos\omega_m t$

For FM, $\omega_i = \omega_c + k_f x(t)$

$= \omega_c + a_m k_f \cos\omega_m t$

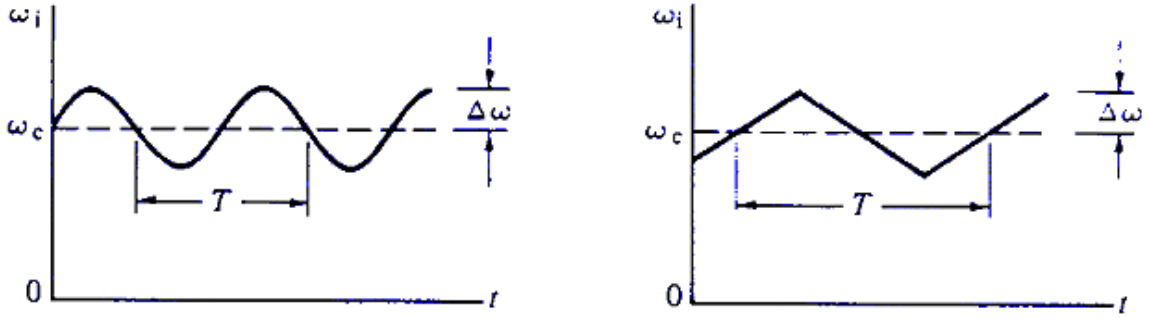
where k_f is the ***frequency modulation constant***; typical units are in **radians per second per volt**.

Define a new constant called the ***peak (maximum) frequency deviation***,

$$\Delta\omega = a_m k_f$$

thus, we have

$$\omega_i = \omega_c + \Delta\omega \cos\omega_m t$$



Angle Modulation: NBFM (Sinusoid)

The phase of this FM signal is

$$\theta(t) = \int_0^t \omega_i(\tau)d\tau + \theta_0$$

$$= \int_0^t [\omega_c + \Delta\omega \cos\omega_m\tau]d\tau + \theta_0$$

$$= \omega_c t + \frac{\Delta\omega}{\omega_m}\sin\omega_m t + 0 \qquad \theta_0 \text{ is set to zero for convenience.}$$

$$= \omega_c t + \beta \sin\omega_m t \qquad \text{where} \quad \boxed{\beta = \frac{\Delta\omega}{\omega_m}}$$

Thus, the resulting FM signal is

$$s_{FM}(t) = \mathrm{Re}\left\{A_c e^{j\theta(t)}\right\}$$

$$= A_c \cos(\omega_c t + \beta \sin\omega_m t)$$

$$= A_c \cos\omega_c t \cos(\beta \sin\omega_m t) - A_c \sin\omega_c t \sin(\beta \sin\omega_m t)$$

For narrowband FM (NBFM) , β is very small so that

$$\cos(\beta \sin\omega_m t) \approx 1, \qquad \sin(\beta \sin\omega_m t) \approx \beta \sin\omega_m t$$

Thus, $\boxed{s_{NBFM}(t) = A_c \cos\omega_c t - A_c \beta \sin\omega_m t \sin\omega_c t}$

The parameter β is called the **modulation index** of the FM signal.

Angle Modulation: NBPM (Sinusoid)

Narrowband PM (NBPM)

Consider $x(t) = a_m \sin \omega_m t$

The phase of this PM signal is

$$\theta(t) = \omega_c t + k_p x(t) + \theta_0$$

$$= \omega_c t + k_p a_m \sin \omega_m t + 0 \qquad \theta_0 \text{ is set to zero for convenience.}$$

$$= \omega_c t + \beta \sin \omega_m t \qquad \boxed{\beta = k_p a_m}$$

The resulting PM signal is

$$s_{PM}(t) = \mathrm{Re}\left\{ A e^{j\theta(t)} \right\}$$

$$= A_c \cos(\omega_c t + k_p a \sin \omega_m t)$$

$$= A_c \cos \omega_c t \cos(\beta \sin \omega_m t) - A_c \sin \omega_c t \sin(\beta \sin \omega_m t)$$

For narrowband PM (NBPM), **β is very small**,

$$\cos(\beta \sin \omega_m t) \approx 1, \qquad \sin(\beta \sin \omega_m t) \approx \beta \sin \omega_m t$$

Thus,

$$\boxed{s_{NBPM}(t) = A_c \cos \omega_c t - A_c \beta \sin \omega_m t \sin \omega_c t}$$

Angle Modulation: NB (Sinusoid)

In summary, if the message signal $x(t)$ is a pure sinusoid, that is,

$$x(t) = \begin{cases} a_m \sin \omega_m t & for \quad PM \\ a_m \cos \omega_m t & for \quad FM \end{cases}$$

Then, $$\phi(t) = \beta \sin \omega_m t$$

where $$\beta = \begin{cases} k_p a_m & \text{for PM} \\ \dfrac{k_f a_m}{\omega_m} = \dfrac{\Delta\omega}{\omega_m} & \text{for FM} \end{cases}$$

$\Delta\omega$: peak frequency deviation

Note that β is known as ***modulation index for angle modulation*** and is **the maximum value of phase deviation for both PM and FM**.

It is **only defined for sinusoidal modulation**.

If $\phi(t)$ **has a bandwidth of** W_B, the **NB angle-modulated signal will have a bandwidth of** $2W_B$.

Angle Modulation: Sinusoid

$$s(t) = A_c \cos(\omega_c t + \beta \sin \omega_m t)$$
$$= A_c \,\mathrm{Re}\left\{ e^{j\omega_c t} e^{j\beta \sin \omega_m t} \right\}$$

The exponential $e^{j\beta \sin \omega_m t}$ is a periodic function of time with a fundamental frequency of ω_m rad/sec. It can be **expanded in a Fourier series**.

$$e^{j\beta \sin \omega_m t} = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_m t} \qquad \text{where} \qquad c_n = \frac{\omega_m}{2\pi} \int_{-\pi/\omega_m}^{\pi/\omega_m} e^{j\beta \sin \omega_m t} e^{-jn\omega_m t} dt$$

Setting $\omega_m t = y$, then $c_n = \dfrac{1}{2\pi} \displaystyle\int_{-\pi}^{\pi} e^{j(\beta \sin y - ny)} dy = J_n(\beta)$

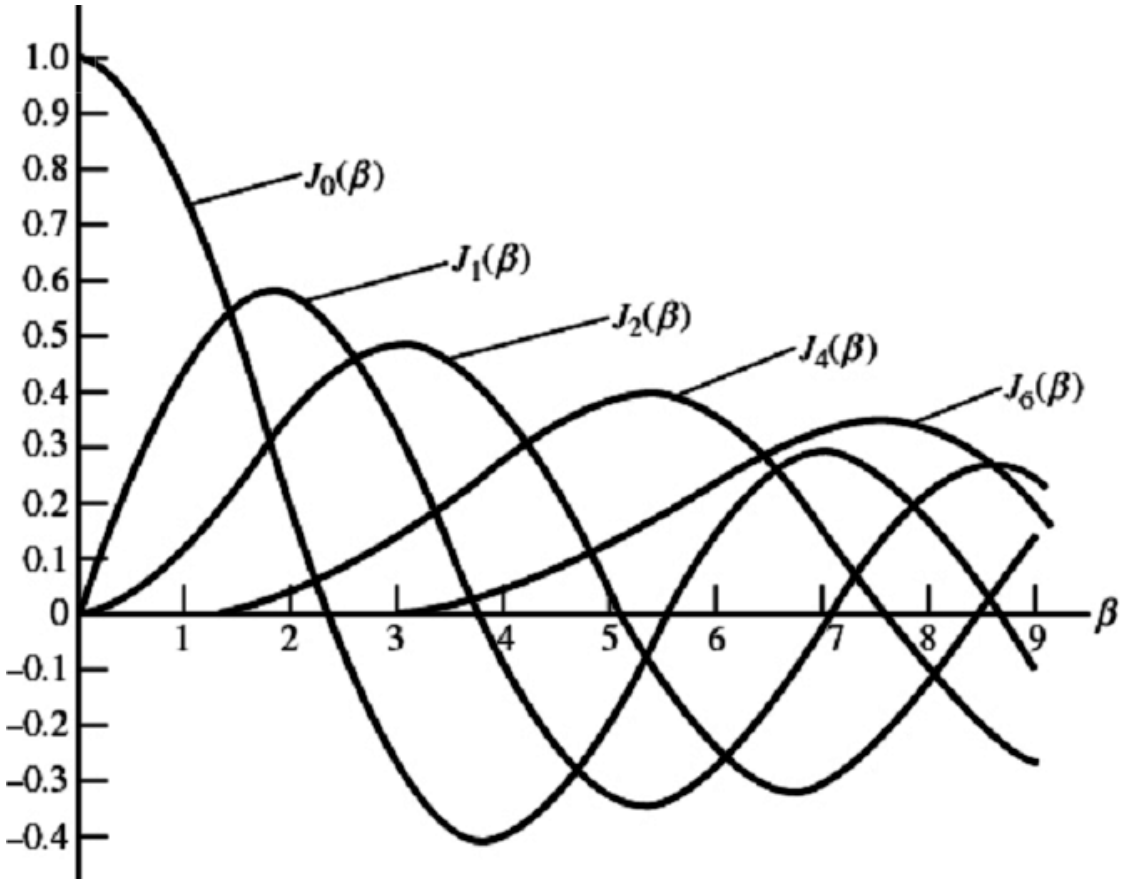
where $J_n(\beta)$ is the **Bessel function of the first kind** of order n and argument β.

$$s(t) = A_c \,\mathrm{Re}\left\{ e^{j\omega_c t} \sum_{n=-\infty}^{\infty} J_n(\beta) e^{jn\omega_m t} \right\}$$
$$= A_c \,\mathrm{Re}\left\{ \sum_{n=-\infty}^{\infty} J_n(\beta) e^{j(\omega_c + n\omega_m)t} \right\}$$

$$\boxed{s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(\omega_c + n\omega_m)t}$$

Angle Modulation: Sinusoid

Bessel function of the first kind of order n and argument β : $J_n(\beta)$



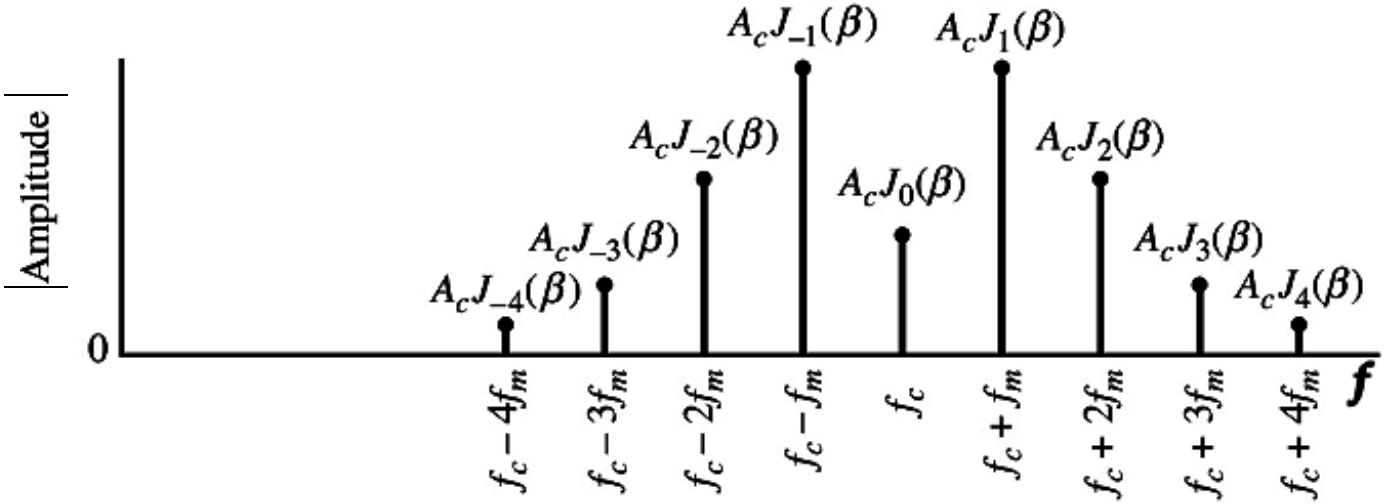
Angle Modulation: Sinusoid

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta)\cos(\omega_c + n\omega_m)t$$

$$J_{-n}(\beta) = \begin{cases} J_n(\beta) & n \text{ even} \\ -J_n(\beta) & n \text{ odd} \end{cases}$$

$$= A_c J_0(\beta)\cos(\omega_c t)$$
$$+ A_c J_1(\beta)\cos(\omega_c + \omega_m)t + A_c J_{-1}(\beta)\cos(\omega_c - \omega_m)t$$
$$+ A_c J_2(\beta)\cos(\omega_c + 2\omega_m)t + A_c J_{-2}(\beta)\cos(\omega_c - 2\omega_m)t$$
$$+ A_c J_3(\beta)\cos(\omega_c + 3\omega_m)t + A_c J_{-3}(\beta)\cos(\omega_c - 3\omega_m)t$$
$$+ \cdots\cdots$$

Spectra



Angle Modulation: Sinusoid

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta)\cos(\omega_c + n\omega_m)t$$

- When β=0 → $J_0(0) = 1$ and all other J_n are zero
  $\Rightarrow$ Only the carrier (no modulation, no sidebands)

- The **spectrum** consists of **carrier-frequency component** plus an **infinite number of sideband components** at frequencies $\omega_c \pm n\omega_m$ (n=1,2,3,…).

- The relative amplitudes of the spectral lines depend on the value of $J_n(\beta)$, and the value of $J_n(\beta)$ becomes very small for large values of n.

- The number of significant spectral lines (that is, having appreciable relative amplitude) is a function of the modulation index β.
  With β **<<1**, only J_0 and J_1 are significant, so the spectrum will consist of carrier and two sideband lines.
  But if β**>>1**, there will be many sideband lines.

Angle Modulation: Sinusoid

Consider the **average power** of $s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta)\cos(\omega_c + n\omega_m)t$

$$\overline{s^2(t)} = A_c^2 \sum_{n=-\infty}^{\infty} J_n^2(\beta)\overline{\cos^2(\omega_c + n\omega_m)t}$$

$$+ A_c^2 \sum_{\substack{n=-\infty \\ m=-\infty \\ n\neq m}}^{\infty} J_m(\beta)J_n(\beta)\overline{\cos(\omega_c + n\omega_m)t\cos(\omega_c + m\omega_m)t}$$

$$= \frac{A_c^2}{2}\sum_{n=-\infty}^{\infty} J_n^2(\beta)$$

$$= \frac{A_c^2}{2} \qquad \because \sum_{n=-\infty}^{\infty} J_n^2(\beta) = 1$$

Thus the **average power dissipated by the angle-modulated signal** over a 1-ohm resistor is

$$P_s = \frac{A_c^2}{2}$$

Angle Modulation: Bandwidth

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta)\cos(\omega_c + n\omega_m)t$$

- The **bandwidth** of the angle-modulated signal with sinusoidal modulation **depends on β and ω_m.**

- **In principle**, the number of sidebands is infinite → the required bandwidth to encompass such a signal is also infinite.

- **In practice**, large portion of the total power is confined to a finite number of sidebands (finite bandwidth)
  → no serious distortion of the signal results if the sidebands outside this bandwidth are lost.

- **Experiments** show that as long as more than **98% of the total signal power** is retained within a certain bandwidth (after a filter/channel), the **distortion is tolerable**.

Angle Modulation: Bandwidth

$$\overline{s^2(t)} = \frac{A_c^2}{2}\sum_{n=-\infty}^{\infty} J_n^2(\beta) = \frac{A_c^2}{2}\left[J_0^2(\beta) + 2\sum_{n=1}^{\infty} J_n^2(\beta)\right] = \frac{A_c^2}{2}\left[J_0^2(\beta) + 2J_1^2(\beta) + 2J_2^2(\beta) + 2J_3^2(\beta) + \cdots\right]$$

Table for Bessel function $J_n(\beta)$

n	$\beta = 0.05$	$\beta = 0.1$	$\beta = 0.2$	$\beta = 0.3$	$\beta = 0.5$	$\beta = 0.7$	$\beta = 1.0$	$\beta = 2.0$	$\beta = 3.0$	$\beta = 5.0$	$\beta = 7.0$	$\beta = 8.0$	$\beta = 10.0$
0	0.999	0.998	0.990	0.978	0.938	0.881	0.765	0.224	−0.260	−0.178	0.300	0.172	−0.246
1	0.025	0.050	0.100	0.148	0.242	0.329	0.440	0.577	0.339	−0.328	−0.005	0.235	0.043
2		0.001	0.005	0.011	0.031	0.059	0.115	0.353	0.486	0.047	−0.301	−0.113	0.255
3				0.001	0.003	0.007	0.020	0.129	0.309	0.365	−0.168	−0.291	0.058
4						0.001	0.002	0.034	0.132	0.391	0.158	−0.105	−0.220
5								0.007	0.043	0.261	0.348	0.186	−0.234
6								0.001	0.011	0.131	0.339	0.338	−0.014
7									0.003	0.053	0.234	0.321	0.217
8										0.018	0.128	0.223	0.318
9										0.006	0.059	0.126	0.292
10										0.001	0.024	0.061	0.207
11											0.008	0.026	0.123
12											0.003	0.010	0.063
13											0.001	0.003	0.029
14												0.001	0.012
15													0.005
16													0.002
17													0.001

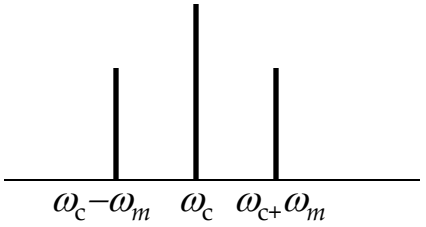
Single underlined: >70% power confined; double underlined: >98% power confined

Angle Modulation: Bandwidth (FM)

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta)\cos(\omega_c + n\omega_m)t \qquad \beta = \frac{\Delta\omega}{\omega_m}$$

For **small β ($\beta << 1$)** only J_0 and J_1 are significant, so the spectrum will consist of carrier and two sideband lines.

$$W_B \approx 2\omega_m$$ **Narrowband FM**



For $\beta \geq 1$, the spectrum, which confines >98% of total power, will consist of carrier and more sideband lines, from the table of Bessel function, we observe $n_{>98\%} \rightarrow (\beta + 1)$

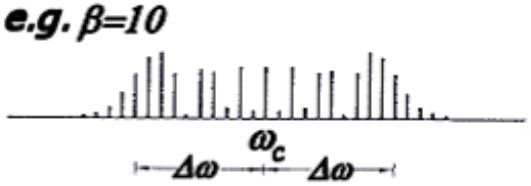
Thus,

$$W_B \approx 2(\beta+1)\omega_m \quad \text{or} \quad W_B \approx 2(\Delta\omega + \omega_m)$$

For **very large β ($\beta >> 1$)**, $W_B \rightarrow 2\beta\omega_m \quad or \quad 2\Delta\omega$

Thus,

$$W_B \approx 2\Delta\omega$$ **Wideband FM**



Angle Modulation: Bandwidth (FM,PM)

Recall:

$$\beta = \begin{cases} k_p a_m & \text{for PM} \\ \dfrac{k_f a_m}{\omega_m} = \dfrac{\Delta\omega}{\omega_m} & \text{for FM} \end{cases}$$

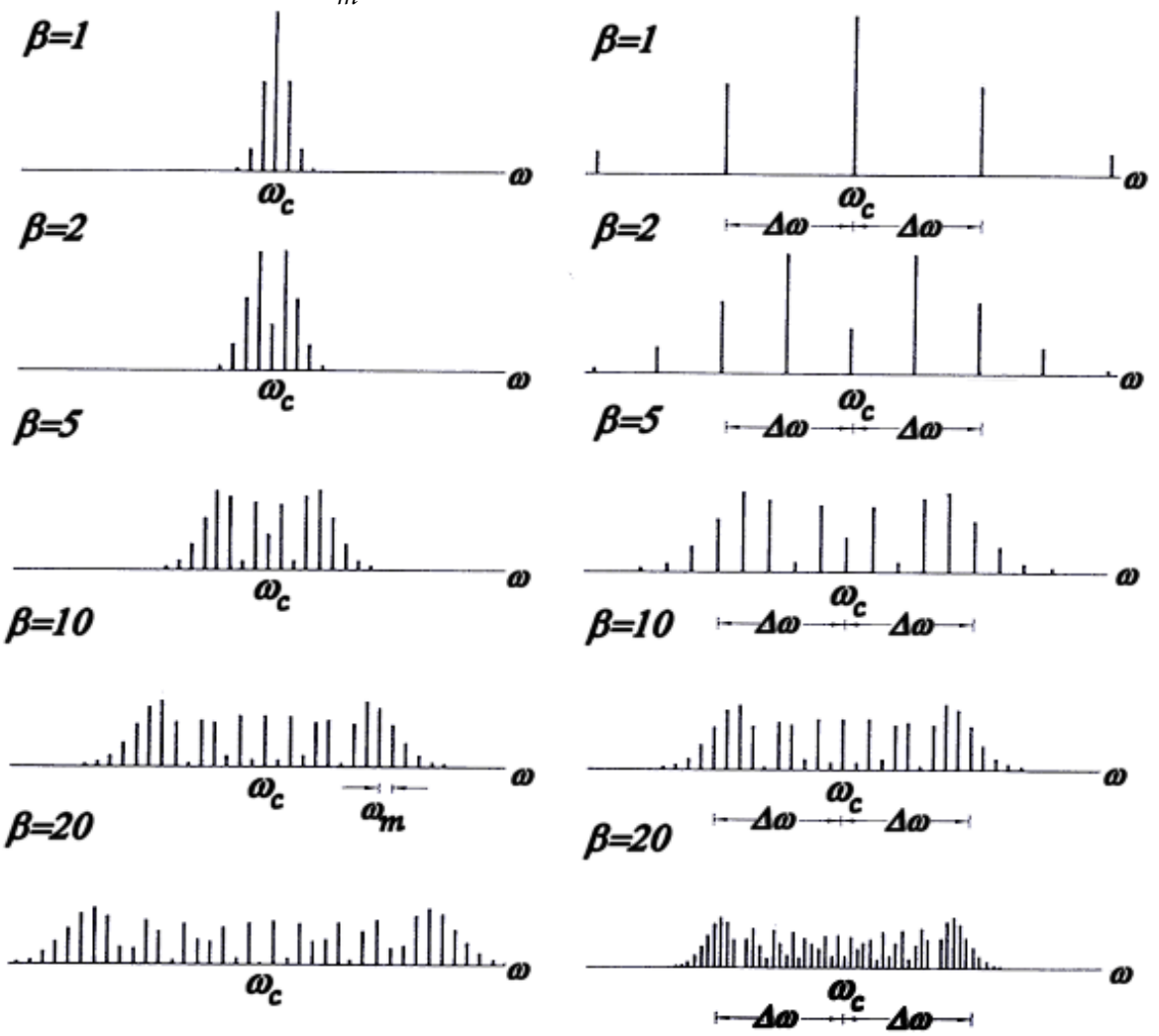
	FM	PM
$\beta << 1$	$W_B \approx 2\omega_m$	$W_B \approx 2\omega_m$
$\beta \geq 1$	$W_B \approx 2(\beta+1)\omega_m$	$W_B \approx 2(\beta+1)\omega_m$
$\beta >> 1$	$W_B \approx 2\Delta\omega$	$W_B \approx 2\beta\omega_m$

Consider a_m = constant, $\beta \geq 1$

$$\omega_m \uparrow \Rightarrow \beta \rightarrow \begin{cases} \text{no change for PM} \\ \downarrow \text{ for FM} \end{cases} ; \; W_B \rightarrow \begin{cases} \uparrow \text{ for PM} \\ \text{slightly} \uparrow \text{ or no change for FM} \end{cases}$$

Angle Modulation: Line Spectra for FM

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta)\cos(\omega_c + n\omega_m)t \qquad \beta = \frac{\Delta\omega}{\omega_m}$$



Angle Modulation: Arbitrary

- The modulation index β is only defined for sinusoidal message signal
- For **arbitrary** message signal $x(t)$, which is bandlimited to ω_M, we define

$$\beta = \frac{\text{peak frequency deviation}}{\text{bandwidth of x(t)}} = \frac{\Delta\omega}{\omega_M}$$

- **Bandwidth**

$$W_B \approx 2(\beta + 1)\omega_M$$

Carson's Rule

- If $\beta << 1$, then $W_B \approx 2\omega_M$
- If $\beta >> 1$, then $W_B \approx 2\Delta\omega$

Angle Modulation: Example

Example: Consider an angle-modulated signal $s(t)=10\cos[(10^8)\pi t + 5\sin 2\pi(10^3)t]$. Find the maximum phase deviation and maximum frequency deviation.

Solution:

$$\theta(t) = \omega_c t + \phi(t) = (10^8)\pi t + 5\sin 2\pi(10^3)t$$

and

$$\phi(t) = 5\sin 2\pi(10^3)t$$

$$\phi'(t) = 5(2\pi)(10^3)\cos 2\pi(10^3)t$$

Thus, maximum phase deviation is $|\phi(t)|_{max}$ = 5 rad;

maximum frequency deviation is $\Delta\omega = |\phi'(t)|_{max} = 5(2\pi)(10^3)$ rad/s
$= 10^4\pi$ rad/s (or Δf=5kHz)

Angle Modulation: Example

Example: Consider an angle-modulated signal $s(t)=10\cos(\omega_c t + 3\sin\omega_m t)$.

Assume PM and f_m= 1kHz, $s_{PM}(t) = A\cos[\omega_c t + k_p x(t)] = 10\cos(\omega_c t + 3\sin\omega_m t)$

With $x(t) = a_m \sin\omega_m t$ → $s_{PM}(t) = 10\cos(\omega_c t + k_p a_m \sin\omega_m t)$

Thus, the modulation index $\beta = k_p a_m = 3$ and bandwidth $W_B \approx 2(\beta+1) f_m = 8kHz$

When f_m is doubled, f_m=2kHz, β=3, and thus bandwidth W_B=2(3+1)2=16kHz

When f_m is halved, f_m=0.5kHz, β=3, and thus bandwidth W_B=2(3+1)0.5=4kHz

Assume FM and f_m=1kHz, $s_{FM}(t) = A\cos[\omega_c t + k_f \int_0^t x(\tau)d\tau] = 10\cos(\omega_c t + 3\sin\omega_m t)$

With $x(t) = a_m \cos\omega_m t$ → $s_{FM}(t) = 10\cos(\omega_c t + \frac{k_f a_m}{\omega_m}\sin\omega_m t)$

Thus, the modulation index $\beta = \frac{k_f a_m}{\omega_m} = 3$ and bandwidth $W_B \approx 2(\beta+1) f_m = 8kHz$

When f_m is doubled, f_m=2kHz, β=3/2, and thus bandwidth W_B=2(3/2+1)2=10kHz

When f_m is halved, f_m=0.5kHz, β=6, and thus bandwidth W_B=2(6+1)0.5=7kHz

Angle Modulation: Generation

Direct Method: - the modulating (message) signal directly controls the carrier frequency

A common method used is to **vary the inductance or capacitance** of a tuned electronic oscillator, e.g. *Voltage-controlled Oscillator (**VCO**).*

Consider an LC circuit, if L_o and C_o are the initial inductance and capacitance, respectively (without any applied signal), the frequency of oscillation is

$$\omega_c = \frac{1}{\sqrt{L_o C_o}}$$

If the capacitance C is varied according to the message signal $x(t)$,
Then $C = C_o + k_f x(t)$; and

$$\omega_s = \frac{1}{\sqrt{L_o C}} = \frac{1}{\sqrt{L_o [C_o + k_f x(t)]}} = \frac{1}{\sqrt{L_o C_o}} \frac{1}{\sqrt{1 + \frac{k_f}{C_o} x(t)}} \approx \omega_c \left(1 - \frac{k_f}{2C_o} x(t)\right)$$

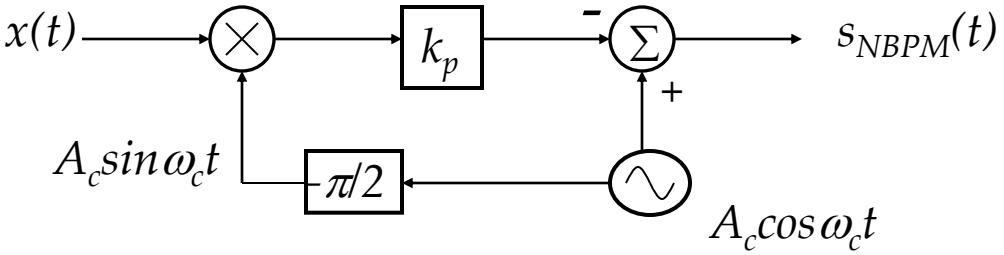
Advantage: large frequency deviation possible
Disadvantage: carrier frequency will drift → need frequency stabilization circuit

DDS techniques

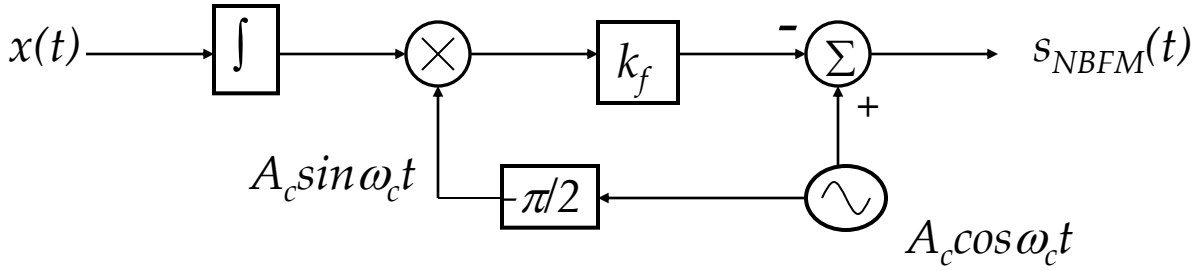
Angle Modulation: Generation

Generation of narrowband angle-modulated signal:

NBPM: $s_{NBPM}(t) \approx A_c \cos\omega_c t - A_c k_p x(t) \sin\omega_c t$



NBFM: $s_{NBFM}(t) \approx A_c \cos\omega_c t - A_c \left[k_f \int_0^t x(\tau) d\tau \right] \sin\omega_c t$



Angle Modulation: Generation

Generation of wideband angle-modulated signal:

Indirect Method: - produce a narrowband angle-modulated signal first;
- convert to wideband signal by using *frequency multiplier*

$s_{NB}(t) = A_c \cos[\omega_c t + \phi(t)]$ → Frequency multiplier → $s_{WB}(t) = A_c \cos[n\omega_c t + n\phi(t)]$

For example, the input-output characteristic of an ideal square-law device is

$$v_o(t) = a v_i^2(t).$$

If the input signal is the FM signal, $v_i(t) = A\cos(\omega_c t + \beta \sin \omega_m t)$,

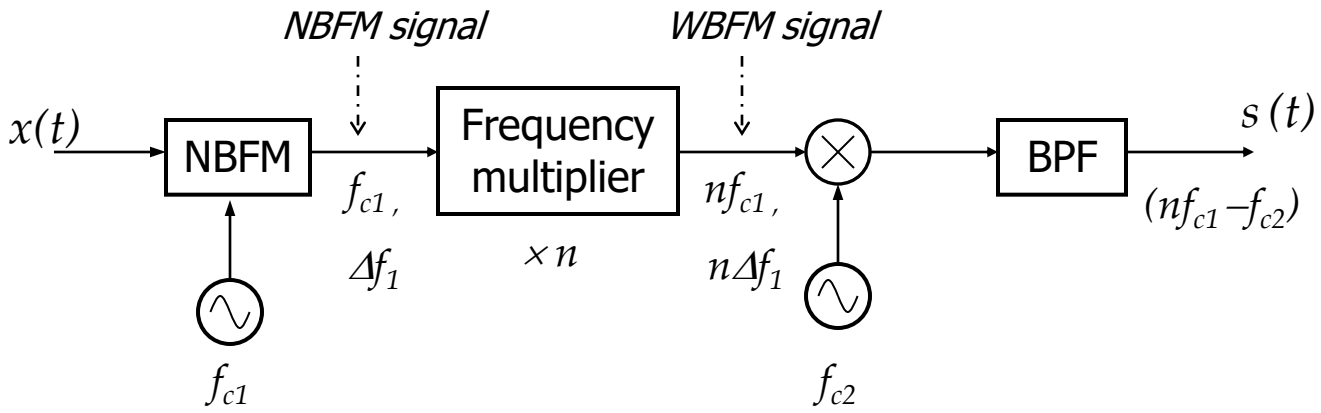
the output is
$$v_o(t) = aA_c^2 \cos^2(\omega_c t + \beta \sin \omega_m t)$$
$$= (1/2)aA_c^2 + \underbrace{(1/2)aA_c^2 \cos(2\omega_c t + 2\beta \sin \omega_m t)}_{\text{frequency doubled signal}}$$

Similarly, use of an nth law device followed by a filter yields a carrier and a modulation index that have been increased by a factor of n.

Angle Modulation: Generation

Use of frequency multiplication increases the carrier of the FM signal as well as the modulation index → very high carrier frequencies

To avoid this, frequency converter is necessary to shift down the carrier frequency.



Angle Modulation: Generation

Example: Consider an NBFM signal passes through a frequency multiplier.

$$s(t) = A\cos(\omega_c t + \beta \sin \omega_m t)$$

With β<0.5 and f_c=200kHz. Let f_m ranges from 50Hz to 15kHz, and let the maximum frequency deviation Δf at the output be 75kHz. Find the required frequency multiplication n and the maximum allowed frequency deviation at the input.

Solution: Using $\beta = \dfrac{\Delta f}{f_m}$, thus

$$\beta_{\min} = \frac{75(10^3)}{15(10^3)} = 5 \qquad \beta_{\max} = \frac{75(10^3)}{50} = 1500$$

If β_1=0.5, where β_1 is the input β, then the required frequency multiplication is

$$n \cdot \beta_1 \le \beta_{\max} \quad \rightarrow \quad n = \frac{\beta_{\max}}{\beta_1} = \frac{1500}{0.5} = 3000$$

The maximum allowed frequency deviation at the input, denoted Δf_1, is

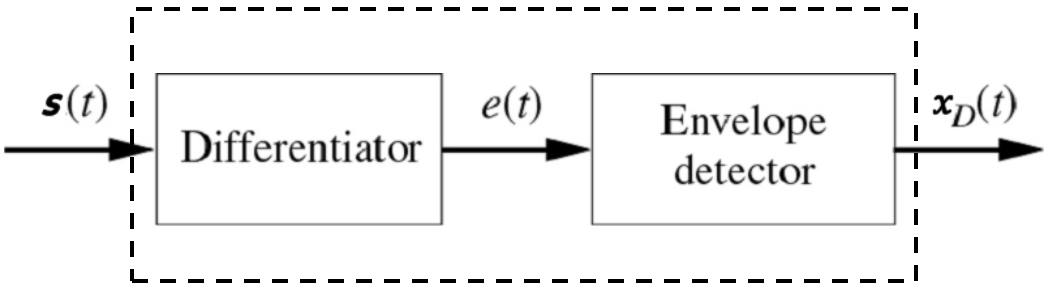
$$\Delta f_1 = \frac{\Delta f}{n} = \frac{75(10^3)}{3000} = 25Hz$$

Angle Modulation: Demodulation

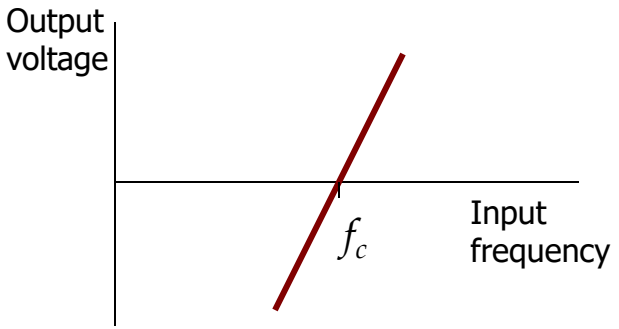
Demodulation: To provide an output signal whose amplitude is linearly proportional to the instantaneous frequency of the input angle-modulated signal.

Direct Method: To use a system that has a linear frequency-to-voltage transfer characteristic.
Such a system is called a **frequency discriminator**.

Conceptually, **differentiator** with a **linear amplitude versus frequency** characteristic can be used as frequency discriminator.



frequency discriminator



Ideal discriminator characteristic

Angle Modulation: Demodulation

For a general FM signal: $s_{FM}(t) = A_c \cos\left[\omega_c t + k_f \int_0^t x(\tau)d\tau\right]$

Assuming that A_c is a constant, we have

$$e(t) = \frac{ds(t)}{dt} = -A_c[\omega_c + k_f x(t)]\sin\left[\omega_c t + k_f \int_0^t x(\tau)d\tau\right]$$

If $k_f x(t) << \omega_c$, it **assembles of an AM signal** whose envelope is

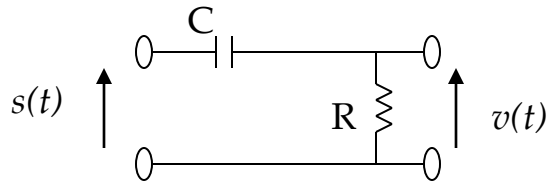
$$x_D(t) = A_c\omega_c\left[1 + \frac{k_f}{\omega_c}x(t)\right]$$

and whose instantaneous angular frequency is $\omega_c + k_f x(t)$

The resulting AM signal can be detected by an envelope detector and the slight variation in the carrier frequency would not be detectable by the envelope detector as long as $k_f x(t) << \omega_c$.

Angle Modulation: Demodulation

Example: An FM signal, $s(t)$, is applied to a RC circuit as shown:



Transfer response of the RC circuit: $H(\omega) = \dfrac{R}{R + 1/(j\omega C)} = \dfrac{j\omega RC}{1 + j\omega RC}$

For $\omega << \dfrac{1}{RC}$; then $H(\omega) \approx j\omega RC$

For the output $v(t) = s(t) * h(t)$

$$V(\omega) = S(\omega) \cdot H(\omega) = S(\omega)\, j\omega RC$$

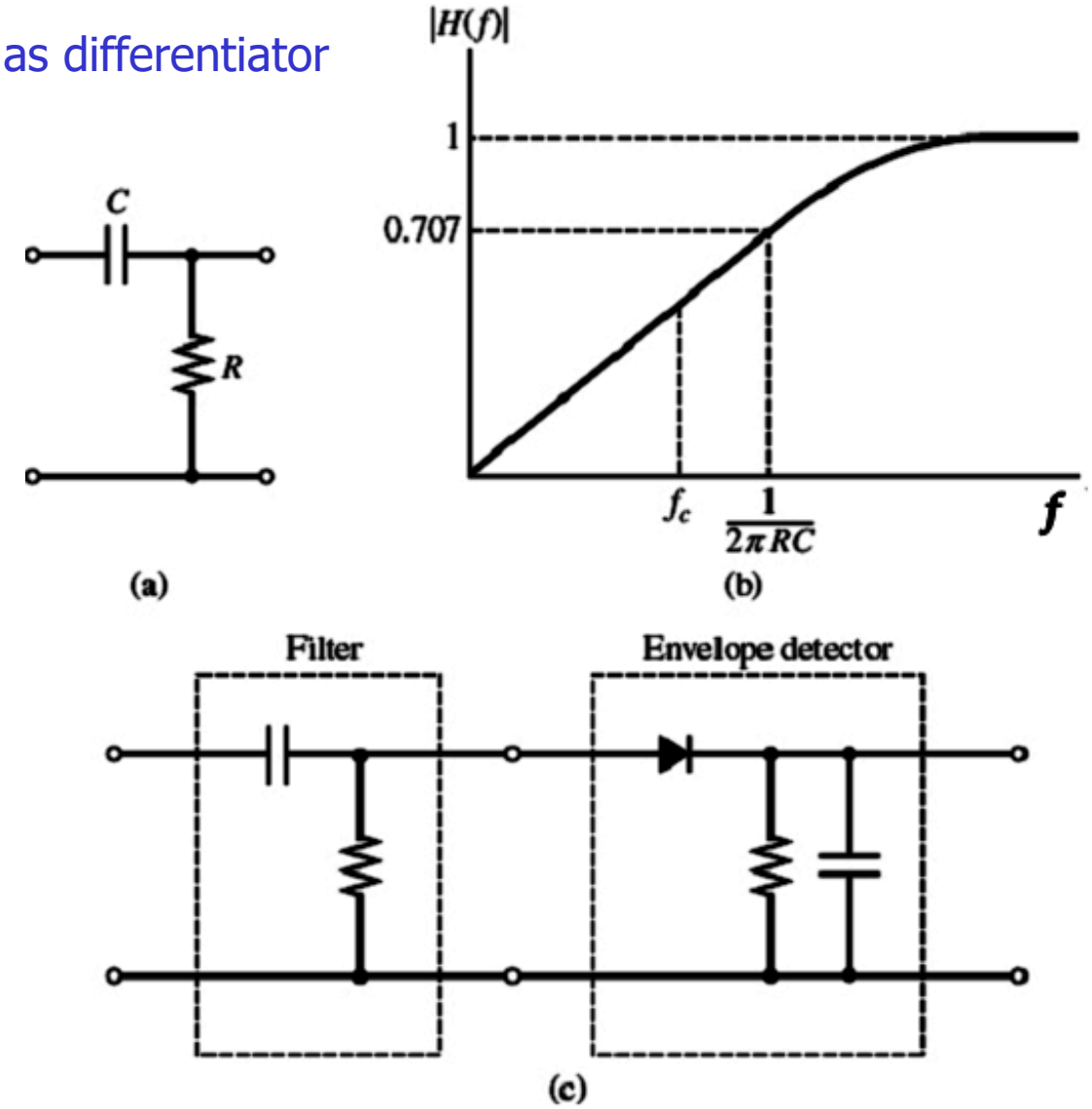
Recall the property of Fourier transform: $\dfrac{df(t)}{d(t)} \leftrightarrow j\omega F(\omega)$

Thus, $v(t) = RC \dfrac{d}{dt}[s(t)]$

So, **the function of this circuit acts as a differentiator and the envelope of the output is proportional to the message signal** (**demodulated**).

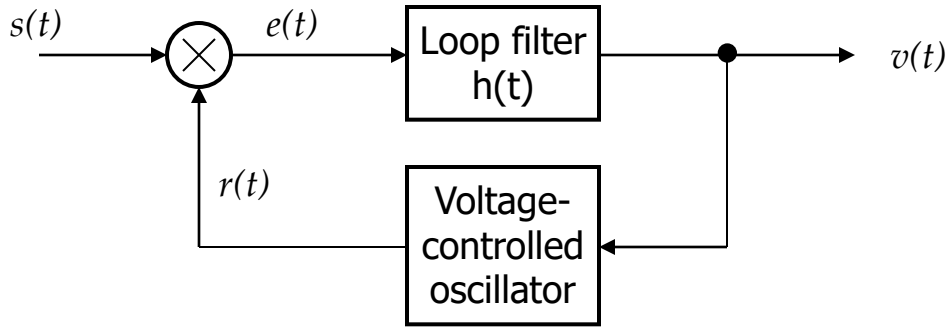
Angle Modulation: Demodulation

RC circuit as differentiator



Angle Modulation: Demodulation

FM Demodulation by Phase-Locked Loop (PLL)



$$s(t) = A_c \cos\left[\omega_c t + \phi_s(t)\right] \qquad where \quad \phi_s(t) = k_f \int_0^t x(\tau)d\tau$$

$$r(t) = A_r \sin\left[\omega_c t + \phi_r(t)\right] \qquad where \quad \phi_r(t) = k_r \int_0^t v(\tau)d\tau$$

$$e(t) = s(t)r(t) = A_c A_r \sin\left[2\omega_c t + \phi_s(t) + \phi_r(t)\right] + \underbrace{A_c A_r \sin\left[\phi_s(t) - \phi_r(t)\right]}_{filter\ output}$$

$$v(t) = \left\{A_c A_r \sin\left[\phi_s(t) - \phi_r(t)\right]\right\} * h(t)$$

$$= \left\{A_c A_r \left[\phi_s(t) - \phi_r(t)\right]\right\} * h(t) \qquad for\ small\ [\phi_s(t) - \phi_r(t)]$$

$$= A_c A_r \phi_d(t) * h(t) \quad where \quad \phi_d(t) = \phi_s(t) - \phi_r(t)$$

Take Fourier transform on both sides,

$$V(f) = A_c A_r \Phi_d(f) H(f)$$

Angle Modulation: Demodulation

$$\phi_d(t) = \phi_s(t) - \phi_r(t) = \phi_s(t) - k_r \int_0^t v(\tau) d\tau$$

$$\frac{d\phi_d(t)}{dt} = \frac{d\phi_s(t)}{dt} - k_r v(t)$$

Take Fourier transform on both sides,

$$j2\pi f \, \Phi_d(f) = j2\pi f \, \Phi_s(f) - k_r V(f)$$

$$\Phi_d(f) = \Phi_s(f) - \frac{k_r}{j2\pi f} V(f)$$

Substitute into $V(f) = A_c A_r \Phi_d(f) H(f)$

gives

$$V(f) = A_c A_r \left[\Phi_s(f) - \frac{k_r}{j2\pi f} V(f) \right] H(f)$$

$$\left[1 + \frac{A_c A_r k_r H(f)}{j2\pi f} \right] V(f) = A_c A_r \Phi_s(f) H(f)$$

$$V(f) = \frac{L(f)}{1 + \frac{k_r L(f)}{j2\pi f}} \Phi_s(f) \quad for \;\; L(f) = A_c A_r k_r H(f)$$

Angle Modulation: Demodulation

$$V(f) = \frac{L(f)}{1 + \frac{k_r L(f)}{j2\pi f}} \Phi_s(f) \quad for \;\; L(f) = A_c A_r k_r H(f)$$

$$V(f) \approx \frac{j2\pi f}{k_r} \Phi_s(f) \quad for \;\; |L(f)| >> 1$$

Take inverse Fourier transform on both sides,

$$v(t) \approx \frac{1}{k_r} \frac{d\phi_s(t)}{dt}$$

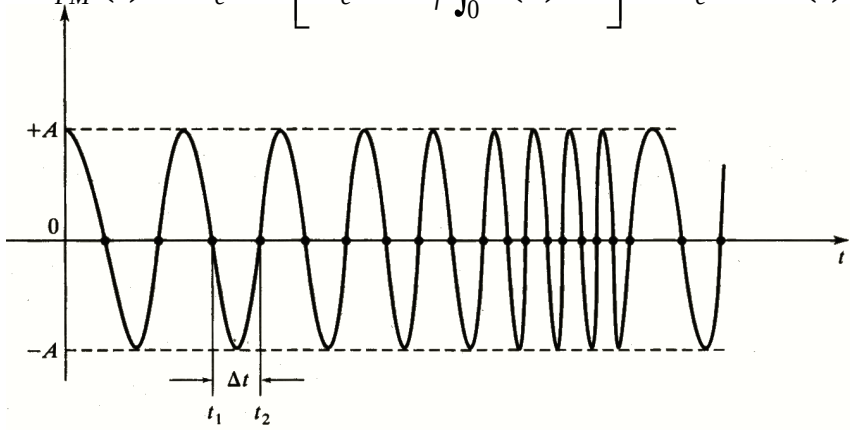
$$v(t) \approx \frac{k_f}{k_r} x(t) \qquad \because \; \phi_s(t) = k_f \int_0^t x(\tau) d\tau$$

So, the **PLL acts as a differentiator** and the output $v(t)$ is proportional to the message signal $x(t)$.

Angle Modulation: Demodulation

FM Demodulation by zero-crossing detector

$$s_{FM}(t) = A_c \cos\left[\omega_c t + k_f \int_0^t x(\tau)d\tau\right] = A_c \cos\theta(t)$$



Let t_1 and t_2 ($t_2 > t_1$) denote the times associated with two adjacent zero crossings of $s(t)$ such that $s(t_1) = s(t_2) = 0$

Thus,

$$\theta(t_2) - \theta(t_1) = \pi = \omega_c(t_2 - t_1) + k_f \int_{t_1}^{t_2} x(\tau)d\tau$$

With $\int_{t_1}^{t_2} x(\tau)d\tau \approx x(t)(t_2 - t_1)$

$$[\omega_c + k_f x(t)](t_2 - t_1) = \pi$$

$$k_f x(t) = \frac{\pi}{\Delta t} - \omega_c$$

Angle Modulation: Demodulation

Let N denote the number of zero crossings of in time T and $t_1, t_2, t_3, \ldots$ denote the times of zero-crossing with $T_1=t_2-t_1$, $T_2=t_3-t_2$, ...
Thus, $T= T_1 + T_2 + \ldots + T_N$

Using the result $$k_f x(t) = \frac{\pi}{\Delta t} - \omega_c$$

gives $$k_f x(t) = \frac{\pi}{T_i} - \omega_c \qquad for\ \ i = 1,2,3,\ldots,N$$

$$T_i = \frac{\pi}{\omega_c + k_f x(t)}$$

$$T = \frac{N\pi}{\omega_c + k_f x(t)}$$

$$k_f x(t) = \frac{\pi N}{T} - \omega_c$$

$$\frac{k_f}{2\pi} x(t) = \frac{N}{2T} - f_c$$

$$\frac{1}{f_c} < T$$

ensures there are zero crossings within time T

$$T << \frac{1}{f_M}$$

ensures no excessive averaging (or smoothing) of $x(t)$

f_M: Bandwidth of $s(t)$

Angle Modulation: FM Receiver

It is similar to the AM superheterodyne receiver with the exception of the addition of a discriminator and possibly a limiter.

Commercial FM radio

